

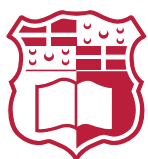
Evaluating the Quality of the LAION-2B Dataset: Insights into Large-Scale Image Dataset Assessment

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Abstract

Generative Artificial Intelligence (AI) systems have enabled the large-scale creation of images, with these technologies becoming increasingly accessible to the public. However, the development of these systems is largely dependent on the existence of massive web-scraped datasets serving as training data, such as Re-LAION-5B, which forms the focus of this research alongside the Stable Diffusion v1.5 model trained on it. Due to the nature of web-scraped training data, there has been increased concern regarding the ethical implications imposed, particularly related to bias and unintended correlations embedded within the training corpus. Specifically, non-semantic or spurious features present within the training dataset may serve as shortcuts for various downstream tasks, negatively impacting model behaviour and fairness outcomes. While these shortcuts can emerge for any task, demographic attributes represent a particularly concerning application, alongside other attributes unrelated to the inherent semantic content of the dataset.

This thesis investigates the demographic bias within the Re-LAION-5B dataset and resultant generative images produced by the Stable Diffusion v1.5 model, with a direct focus on identifying spurious features within these two components and determining whether the generative model amplifies, alters, or mitigates those spurious features identified within the respective training corpus. The study adopts a classifier-based approach derived from prior research, in which discriminative models are trained to distinguish between Re-LAION-5B and Stable Diffusion v1.5 generated images. These classifiers utilise images that have undergone specific transformations. These transformations systematically reduce the semantic information available to the model and force reliance on specific low-level image features, thereby exposing the presence of dataset specific spurious patterns that can be exploited by the model for image source identification.

Building upon this classifier based analysis, the relevant image sets used in the discrimination task are further annotated in terms of gender and skin-tone attributes. This secondary analysis investigates whether the previously identified dataset specific spurious features demonstrate correlations with these sensitive demographic attributes, allowing for the detection of harmful features that may serve as model shortcuts for deriving gender or skin-tone classifications. To further contextualise these findings and assess the real world implications of observed biases, the demographic distributions of retrieved and generated images across 200 occupation categories are compared against real world demographic statistics, revealing patterns of representation bias and potential amplification effects in the generative model.

Finally, based on the spurious features identified in the prior sections, this thesis

outlines a comprehensive set of inference-time strategies for debiasing the Stable Diffusion v1.5 model without requiring full model retraining. These proposed methods aim to mitigate the effect that identified spurious features have on resultant generated images, offering practical approaches for improving fairness in deployed systems. By explicitly linking dataset characteristics, spurious feature learning mechanisms, and generative bias outcomes, this work contributes both methodological insights for evaluating large-scale multimodal datasets and practical interventions for the development of fairer text-to-image generation systems.

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1 Introduction

1.1 Problem Definition

The rapid advancement of text-to-image generative models has fundamentally transformed visual content creation, with diffusion-based architectures achieving unprecedented image quality and text-prompt controllability. Among these, Stable Diffusion [1] has emerged as a particularly influential model due to its open-source availability, widespread deployment across commercial and research applications, and the extensive ecosystem of tools and fine-tuned variants built upon it.

These systems rely critically on massive web-scraped datasets to learn the mapping between textual descriptions and visual content. Re-LAION-5B, containing over 5 billion image-text pairs harvested from the internet, represents one of the largest publicly available training corpora for such models [2]. However, the scale and automated nature of web scraping introduces systematic biases that reflect and potentially amplify existing societal inequities present in online content [3].

While demographic bias in AI systems is well-documented across domains such as natural language processing [4], machine learning systems [5], and computer vision [6], there exists a gap between detecting bias in generative model outputs and understanding the specific low-level image features and dataset characteristics that cause and perpetuate these biases.

Current research predominantly adopts an outcome-focused perspective, measuring demographic distributions in generated images and comparing them against desired benchmarks [7, 8]. This approach treats models as black boxes, diagnosing the symptoms of bias without investigating the underlying representational features that enable biased behavior.

A fundamental yet underexplored dimension of this problem involves spurious features—low-level, non-semantic image characteristics including color statistics, texture patterns, edge distributions, frequency domain properties, and compression artifacts [9, 10]. Unlike semantic features that capture meaningful visual content, spurious features represent superficial statistical regularities that can enable biased shortcuts across downstream tasks. Critically, when these spurious features correlate with sensitive demographic attributes, models can make predictions based on superficial cues rather than semantic understanding, creating an insidious form of bias that operates beneath standard evaluation metrics. Furthermore, it remains unclear how generative training processes transform these spurious patterns, making it difficult to predict how training data biases will manifest in deployed systems.

Existing bias mitigation approaches typically require expensive interventions:

curating balanced training datasets [11], implementing fairness-constrained optimization during training [12], or fine-tuning models on carefully constructed demographic distributions [13]. These methods demand substantial computational resources and extensive human annotation, creating barriers for researchers and practitioners working with pre-trained models. Furthermore, outcome-focused mitigation strategies may successfully balance demographic distributions without addressing the underlying spurious features that enable biased shortcuts, potentially leading to superficial corrections that fail to generalize across contexts or evaluation metrics.

This thesis addresses these gaps through a mechanistic investigation of spurious features in the Re-LAION-5B dataset and Stable Diffusion v1.5 model. Rather than solely measuring demographic outcomes, this research applies established methodologies from prior work to expose and characterize the low-level image features that distinguish dataset sources and correlate with sensitive attributes. By understanding which specific spurious patterns enable biased behavior in this influential dataset-model pair, this work provides empirical evidence regarding bias mechanisms and evaluates targeted inference-time interventions that address root causes rather than symptoms.

1.2 Motivation

The main motivating factor is the critical role that spurious features within generative models play in the generative AI ecosystem and the urgent need for accessible fairness interventions.

Influence and Impact: Re-LAION-5B serves as training data for multiple widely-deployed text-to-image models, while Stable Diffusion v1.5 has achieved widespread adoption with millions of users generating billions of images through platforms like DreamStudio, Hugging Face, and numerous third-party integrations. Despite this pervasive influence, comprehensive analysis of the spurious features within Re-LAION-5B and their propagation through Stable Diffusion v1.5 remains limited. Understanding bias mechanisms in these specific systems directly impacts the fairness of deployed generative AI at scale.

Computational Accessibility: Training or fine-tuning large-scale diffusion models requires substantial computational resources placing such interventions beyond reach for most researchers and practitioners. By focusing on inference-time methods applicable to frozen models, this research demonstrates fairness interventions accessible to smaller research groups, independent developers, and organizations with limited computational budgets who cannot implement training-time solutions.

Reproducibility and Transparency: Much fairness research examines proprietary systems where training data and model weights remain inaccessible, limiting independent verification. The open-source nature of Stable Diffusion v1.5 and the research availability of Re-LAION-5B enable findings to be independently verified, replicated, and extended by the broader community, strengthening the credibility and impact of this work.

Beyond Outcome Measurement: While existing research predominantly measures demographic distributions in generated images, understanding *why* biases emerge—specifically, which low-level features enable biased shortcuts—enables interventions that address root causes rather than symptoms. This mechanistic understanding is essential for developing robust fairness solutions that generalize across contexts rather than superficial corrections that fail when evaluation metrics or use cases change.

1.3 Aims and Objectives

The overarching aim of this thesis is to apply established spurious feature detection methodologies to the Re-LAION-5B dataset and Stable Diffusion v1.5 model, systematically characterise the spurious features present in each, quantify their correlation with sensitive demographic attributes, and evaluate the effectiveness of existing inference-time debiasing techniques when applied to this specific dataset-model pair.

To accomplish this aim, the research pursues the following specific objectives:

Objective 1: Dataset Acquisition, Preparation, and Annotation

Retrieve images from Re-LAION-5B and generate corresponding images using Stable Diffusion v1.5 across a pre-defined set of occupation categories to enable systematic bias analysis. This includes addressing technical challenges such as deriving image embeddings in the absence of pre-computed CLIP embeddings from the Re-LAION-5B dataset. Annotate all retrieved and generated images for gender and skin-tone attributes using established computer vision models to enable demographic analysis and spurious feature correlation assessment.

Objective 2: Identify and Implement Spurious Feature Detection Methods

Identify and implement established classifier-based methodologies for detecting spurious features that distinguish between dataset sources. Apply systematic image transformations (color extraction, edge maps, frequency filtering) to isolate non-semantic features and train discriminative classifiers to characterise dataset-specific spurious patterns in Re-LAION-5B versus Stable Diffusion v1.5 outputs.

Objective 3: Assess Spurious Feature Correlations with Demographics

Analyze whether identified spurious features exhibit systematic correlations with sensitive demographic attributes (gender and skin-tone) within both Re-LAION-5B and Stable Diffusion v1.5 generated images. Quantify these correlations to determine the extent to which spurious features enable demographic classification shortcuts, and compare demographic distributions across occupations against real-world statistics to identify representation biases.

Objective 4: Identify and Implement Inference-Time Debiasing Methods

Identify and implement established inference-time debiasing techniques applicable to Stable Diffusion v1.5 without model retraining. Adapt these methods based on identified spurious features and evaluate their effectiveness through multiple metrics including demographic representation balance, spurious feature mitigation, and maintenance of image quality. Assess trade-offs between fairness improvements and generation quality to inform practical deployment guidelines.

Objective 5: Derive Conclusions on Bias Mechanisms and Mitigation Effectiveness

Synthesize findings to characterize how spurious features manifest in Re-LAION-5B and propagate through Stable Diffusion v1.5, determining whether the generative process preserves, amplifies, or mitigates training data biases. Document the effectiveness of inference-time debiasing methods on a widely-deployed production model and provide evidence-based recommendations for practitioners seeking to deploy fairer text-to-image generation systems.

Through these objectives, this thesis provides empirical analysis of spurious features and demographic bias in Re-LAION-5B and Stable Diffusion v1.5, demonstrating the practical application of existing methodologies to real-world systems and contributing actionable insights for developing more equitable text-to-image generation technologies.

1.4 Document Structure

2 Background & Literature Review

2.1 Re-LAION-5B Dataset

Re-LAION-5B is a large-scale, openly released image-text dataset comprising approximately 5.85 billion image-text pairs harvested from the publicly accessible portions of the internet via Common Crawl, a non-profit organisation that continuously indexes web content and makes its archived crawl data freely available to researchers [2, 14]. The dataset is a re-release of the original LAION-5B, itself introduced by Schuhmann et al. [2] as one of the largest publicly available multimodal training corpora at the time of its release. The re-release was motivated by the discovery of CSAM within the original corpus, with Re-LAION-5B representing a re-filtered version produced through improved safety filtering pipelines, including enhanced perceptual hashing and classifier-based detection methods designed to remove harmful content [15].

The construction pipeline of the original LAION-5B, which Re-LAION-5B is derived from, is structured as follows. HTML pages indexed by Common Crawl are parsed to extract image URLs paired with their associated alt text, yielding raw image-caption pairs at web scale. These pairs are subsequently filtered using a CLIP-based semantic similarity score [16], computed as the cosine similarity between image and text embeddings using OpenAI’s ViT-B/32 CLIP model for English pairs and a multilingual CLIP variant for all others. As described by Schuhmann et al. [Schuhmann_2022_LAION5B], pairs below a cosine similarity of 0.28 were discarded for English content and below 0.26 for non-English content, a step that removed approximately 90% of the original 50 billion candidate pairs. Additional filtering stages remove near-duplicate pairs using locality-sensitive hashing, apply NSFW classifiers to exclude explicit content, and enforce minimum resolution and caption length requirements to discard low-quality samples.

The resulting dataset is stratified by detected language into three subsets: LAION-2B-en, comprising approximately 2.32 billion English-language image-text pairs; LAION-2B-multi, comprising approximately 2.26 billion pairs spanning over 100 other languages; and LAION-1B-nolang, comprising approximately 1.27 billion samples for which no language could be reliably detected, which frequently correspond to images of products, places, or content dominated by proper nouns [Schuhmann_2022_LAION5B]. Together these three subsets constitute the full LAION-5B corpus of approximately 5.85 billion pairs.

A further aesthetics-filtered derivative was constructed exclusively from LAION-2B-en. Using a linear classifier trained on CLIP ViT-L/14 image embeddings to

predict human aesthetic ratings, Schuhmann [laionAesthetics] produced a series of nested subsets of LAION-2B-en filtered by predicted aesthetic score. The subset used for training Stable Diffusion v1.5, referred to as LAION-Aesthetics v2 5+, retains only those pairs with a predicted aesthetic score exceeding 5.0 out of 10, an original image resolution of at least 512×512 pixels, and an estimated watermark probability below 0.5 [sd15modelcard], yielding approximately 600 million image-text pairs. LAION-Aesthetics is therefore not a separate corpus but a quality-filtered offshoot of the English-language component of LAION-5B.

Despite the scale and ambition of the dataset, several significant concerns have been raised regarding its content. Birhane et al. [3] conducted an audit of LAION-400M, a predecessor to LAION-5B, and identified substantial quantities of misogynistic, pornographic, and racially derogatory content embedded within the corpus, reflecting the unfiltered nature of web-scraped data. These findings were later extended to LAION-5B itself, reinforcing concerns that automated safety filtering is insufficient to remove harmful material at billion-scale. Beyond explicit harmful content, the web-scraped construction process introduces systematic demographic and representational imbalances, as the statistical composition of online imagery reflects existing societal inequities in content production, platform access, and visual representation [3, 5].

A practically significant limitation of the Re-LAION-5B release concerns the availability of pre-computed embeddings. The original LAION-5B release included pre-computed CLIP image and text embeddings alongside image URLs and captions, enabling efficient nearest-neighbour retrieval and semantic search without requiring users to independently process raw images. These embeddings underpinned tools such as the LAION CLIP retrieval service, which allowed researchers to query the dataset semantically. In the Re-LAION-5B re-release, these pre-computed embeddings were not made publicly available, as the re-filtering process altered the dataset composition and the associated embedding infrastructure was not reconstructed at the time of release [15]. As a result, any research requiring embedding-based retrieval from Re-LAION-5B must independently compute CLIP embeddings from raw image downloads, introducing substantial computational overhead and making large-scale semantic queries considerably more difficult to execute than was the case with the original release. For this thesis, this absence necessitates the derivation of image embeddings as part of the data acquisition pipeline.

2.2 Spurious Features in Machine Learning

Defining Spurious Features and Spurious Correlations

A key challenge in deploying machine learning models reliably is their tendency to learn features that are predictive within the training data but carry no causal or semantic relationship to the target variable. Such features are termed *spurious features*, and the statistical associations they form with target labels are known as *spurious correlations* [17, 18]. Formally, a spurious correlation exists when a feature f co-occurs with a label y across the training distribution $p_{\text{train}}(f, y)$ but this co-occurrence does not persist under distribution shift, i.e., $p_{\text{train}}(f | y) \neq p_{\text{test}}(f | y)$, or when f carries no causal information about y despite being statistically correlated with it. A canonical illustration is a gender classifier trained on a dataset where women are disproportionately photographed wearing dresses: the model may learn clothing type as a reliable decision cue rather than facial or morphological features. This correlation holds within the training distribution but breaks under shift, for instance when the model encounters images of men in kilts or women in suits, exposing the feature as spurious. A model that learns to exploit such a feature in lieu of causally relevant ones is said to learn a *shortcut* [19].

Geirhos et al. [19] provide a comprehensive treatment of shortcut learning in the deep learning literature. They characterise a shortcut as any decision rule that performs well on the training distribution by exploiting a feature correlated with the correct label but which does not reflect the underlying task concept. Shortcuts are distinguished from genuinely useful features not by their statistical properties within the training set but by their failure to generalise when the spurious correlation is broken, either through natural distribution shift or deliberate intervention. Critically, Geirhos et al. argue that shortcut learning is not a pathological edge case but a rational consequence of empirical risk minimisation: given a choice between a reliable spurious cue and a harder-to-learn causal feature, gradient-based optimisation will consistently favour whichever feature reduces training loss most efficiently.

Taxonomies of Spurious Features in Computer Vision

Spurious features in visual data can be broadly categorised along two dimensions: the level of the image hierarchy at which they reside, and whether they reflect properties of the subject, the context, or the acquisition process.

At the *pixel and statistical level*, low-level image attributes encompassing mean colour channel values, brightness distributions, contrast statistics, and jpeg compression artefacts can serve as reliable proxies for class labels, thereby enabling

shortcut learning particularly when the dataset collection process introduces systematic correlations between acquisition conditions and class membership. Jo and Bengio [20] demonstrate that Convolutional Neural Networks (CNN) trained on benchmark datasets are inclined to learn surface statistical regularities as opposed to high-level abstractions, achieving high classification accuracy even when these regularities bear no semantic relationship to the class concept. Furthermore, Geirhos et al. [21] demonstrate how ImageNet-trained CNNs exhibit a strong *texture bias*, opting to rely on texture for object classification as opposed to shape, an effect particularly apparent when the texture of one object is rendered onto the shape of another.

At the *frequency domain level*, the Fourier decomposition of images separates low-frequency components, such as smooth colour gradients and global illumination, from high-frequency components including edges, fine textures, and compression ringing. Wang et al. [22] demonstrate that CNNs can exploit high-frequency features imperceptible to the human eye as spurious shortcuts, and that adversarially introduced high-frequency perturbations can substantially alter model decisions without any visible change to image content. This sensitivity is particularly concerning in datasets assembled from diverse web sources, where different image origins often carry systematically different frequency signatures arising from variation in camera hardware, processing pipelines, and compression settings.

At the *contextual and background level*, spurious features can arise from statistical regularities in scene composition rather than in the subject itself. Beery et al. [23] demonstrate this in a wildlife recognition setting, showing that classifiers trained on images where animals consistently appear against certain background types fail severely when the same animal appears against an unfamiliar background, revealing that the background has been learned as a more reliable shortcut than animal morphology. Xiao et al. [24] reach similar conclusions in the context of ImageNet, demonstrating that models achieve non-trivial classification accuracy when trained and tested on background-only images with foreground objects masked, establishing that contextual scene statistics constitute a pervasive spurious signal in standard benchmarks. Torralba and Efros [25] earlier identified a related phenomenon at the dataset level: models trained on one visual dataset and evaluated on another exhibit sharp performance drops, demonstrating that much of what models learn reflects dataset-specific statistical regularities rather than generalisable visual concepts.

At the *semantic and co-occurrence level*, spurious correlations can be encoded in higher-level statistical regularities such as the tendency for certain object categories to co-occur with certain scenes, or for certain demographic attributes to be associated with particular activities. These correlations are not artefacts of image acquisition but reflect genuine societal patterns or collection biases that are absorbed into the training distribution and subsequently exploited by the model. Zhao et al. [26] demonstrate

that gender-stereotypical object co-occurrences in image captioning datasets cause models to amplify those associations in their outputs, with activities such as cooking and shopping disproportionately paired with female subjects and sports with male subjects, associations that persist and intensify in model predictions beyond their frequency in the training data. Singh et al. [**Singh_2020_DontJudgeObject**] further demonstrate the consequences of such co-occurrence dependencies, showing that models trained on standard benchmarks rely heavily on the presence of contextually expected objects and scenes, and that performance degrades substantially when objects appear without their typical co-occurring context, confirming that semantic co-occurrence regularities function as learned shortcuts rather than robust recognition features.

At the *structural and geometric level*, the shapes and spatial arrangements of objects constitute a further tier of potentially spurious structure that is independent of both global colour statistics and semantic object identity. Classical edge detection algorithms such as the Canny edge detector [**Canny_1986_ComputationalApproachEdgeDetection**] reduce an image to a binary boundary map that captures object contours while discarding colour and texture entirely, preserving only structural information about object shape and boundary distributions. Class-agnostic segmentation tools such as the Segment Anything Model [**Kirillov_2023_SegmentAnything**] extend this to cleaner, higher-quality contour representations that are minimally affected by local pixel variation. Monocular depth estimation encodes the relative spatial geometry of a scene, capturing information about object positioning and scene layout without reference to object semantics or surface appearance. Since these structural properties — edge distributions, object silhouettes, and scene geometry — are systematically shaped by the contexts and environments in which images are captured, they can carry dataset-specific or demographic-specific spurious signals that are entirely invisible to pixel-level or semantic-level analysis. The sufficiency of local spatial structure for classification is further evidenced by Brendel and Bethge [**Brendel_2019_BagNet**], who demonstrate that models operating solely on small independent image patches of fixed size, with no access to global spatial relationships, achieve surprisingly high accuracy on ImageNet, confirming that local structural features within small receptive fields are sufficient to support above-chance classification without any holistic understanding of image content.

At the *person level*, in datasets where the target variable relates to people, the body introduces a distinct layer of spurious signals that compounds the background-level effects described above. A person’s appearance, shape, spatial location, and pose may each independently correlate with demographic group membership when the collection conditions of a dataset introduce systematic

differences in how different groups are photographed. This has direct consequences for a class of bias mitigation approaches known as *fairness through blindness*, which attempt to remove protected attribute information by occluding the person in an image [Hendricks_2018_WomenAlsoSnowboard, Wang_2019_BalancedDatasetsNotEnough]. Because person-level spurious features are intertwined with background signals and encoded across multiple representational layers simultaneously, such occlusion-based interventions may leave substantial demographic information intact, a limitation examined in detail in Section 2.2.

At the *semantic representation level*, richer image representations that preserve object identity while discarding pixel-level detail can themselves carry dataset-specific spurious signals across a hierarchy of abstraction. Semantic segmentation maps, which assign a class label to every pixel, retain the spatial distribution of scene categories while removing texture and colour, encoding information about which types of environments and objects are systematically associated with which classes or groups. Object detection representations, which reduce each image to a set of labelled bounding boxes, preserve object co-occurrence statistics and coarse spatial layout while discarding all internal pixel content. Image captions produced by vision-language models discard all visual information in favour of natural language but still encode semantic regularities that reflect the statistical composition of the underlying image distribution. VAE reconstructions, which pass images through a learned bottleneck to compress and regenerate visual content — as implemented in latent diffusion architectures such as that of Rombach et al. [1] — provide a further diagnostic: if reconstructed images remain discriminable after compression, the bias resides primarily in semantic content rather than in low-level acquisition artefacts such as jpeg compression or sensor noise. Taken together, these representation types demonstrate that spurious features are not confined to the pixel domain but permeate the full hierarchy of visual abstraction.

Spurious Features and Demographic Bias

The intersection of spurious features and demographic fairness constitutes a particularly consequential research area. When a spurious feature correlates not merely with a label but with a sensitive attribute such as gender, race, or skin tone, a model that exploits the shortcut will systematically produce disparate outcomes across demographic groups. Buolamwini and Gebru [6] provide a foundational empirical demonstration of this: commercial gender classifiers trained on datasets overrepresenting lighter-skinned individuals exhibit substantially higher error rates for darker-skinned individuals, a disparity attributable in part to the model relying on skin-tone correlates as classification shortcuts. Mehrabi et al. [5] offer a

comprehensive survey cataloguing the mechanisms by which such demographic shortcuts enter models, the most relevant of which being representation bias (some groups being underrepresented in training data), measurement bias (features being measured differently across groups) and aggregation bias (heterogeneous sub-populations being pooled under a single model).

Critically, even features that appear neutral — colour statistics, pose distributions, background scenes — can function as proxies for demographic attributes when the dataset composition creates systematic correlations between these low-level properties and group membership. Meister et al. [9] term these *gender artifacts* and demonstrate that they are effectively impossible to fully remove from standard visual datasets without also removing information necessary for downstream tasks. This proxy mechanism means that standard debiasing interventions which remove explicit demographic signals may leave intact the spurious low-level features that encode equivalent information, producing only superficial corrections.

Spurious Features in Text-to-Image Generative Systems

The concern about spurious features takes on a distinct character in the context of text-to-image generative models. Whereas discriminative models learn to *exploit* spurious correlations to make predictions, generative models trained on biased data learn to *reproduce* those correlations in their outputs: biased statistical regularities present in the training corpus become embedded in the model’s learned prior over images, causing the model to synthesise spurious feature patterns even when the conditioning prompt contains no information that should elicit them.

The training pipeline of latent diffusion models such as Stable Diffusion [1] illustrates this concretely. The model learns a joint distribution over image latents conditioned on text embeddings derived from CLIP [16]. Since CLIP itself is pretrained on web-scraped image-text pairs, it encodes the same societal and compositional biases present in internet imagery [2, 3]. Spurious correlations in the training data between text tokens (e.g. occupation words) and low-level image features (e.g. skin tone distributions, colour palettes, or compositional conventions) are absorbed into the diffusion model’s score function, causing those features to be hallucinated at generation time even in the absence of explicit demographic prompting.

Bianchi et al. [8] demonstrate this empirically, showing that text-to-image models not only reflect the demographic under-representation present in their training data but actively amplify stereotypical associations: prompts specifying occupations produce generated images with demographic distributions more extreme than those documented in real-world labour statistics. Luccioni et al. [7] extend this finding across a broader set of models and prompt categories, confirming that the amplification effect

is systemic rather than model-specific. Critically, neither study addresses the question of *which specific low-level image features* encode these biases, treating the model as a black box and measuring only demographic outcomes.

Understanding Bias in Large-Scale Visual Datasets

Zeng et al. [10] aim to identify the concrete forms of bias that make large-scale pre-training datasets easily distinguishable by neural networks. Their work builds on the dataset classification task introduced by Torralba and Efros [25] in 2011 and revisited by Liu and He [Liu_2024_DecadeBattleDatasetBias] in 2024, who found that three widely-used datasets — YFCC100M [Thomee_2016_YFCC100M], CC12M [Changpinyo_2021_CC12M], and DataComp-1B [Gadre_2023_DataComp] — can be classified by a neural network with 84.7 % accuracy. While this confirms the datasets are biased, it does not explain *which* image properties are responsible. Zeng et al. address this gap.

Transformation-Based Bias Quantification

The central contribution is a transformation-based framework. The idea is straightforward: apply a transformation that preserves one type of visual information while removing all others, train a dataset classifier on the transformed images, and use the resulting accuracy to measure how much bias that type of information carries. The three datasets — YFCC100M, CC12M, and DataComp-1B, referred to collectively as YCD — represent the kind of large-scale internet-scraped data used to pre-train modern vision models. For each transformation, 1M images per dataset are used for training and 10K for validation using a ConvNeXt-Tiny model trained for 30 epochs. On unmodified images, Zeng et al. achieve a reference accuracy of 82.0 %, which all transformation results are compared against.

The transformations cover the full range of visual information from raw pixels to high-level semantics. At the semantic level, segmentation using ViT-Adapter-Large with the ADE20K vocabulary (150 classes) converts each image into a colour-coded class map, removing texture but keeping the spatial layout of objects; the classifier reaches 69.8 % accuracy. Object detection using ViTDet-Huge with the LVIS vocabulary (1,203 classes) keeps bounding boxes and class labels but removes all pixel content, yielding 61.9 %. Image captioning with LLaVA 1.5 removes all visual information entirely, yet short captions still achieve 63.8 % and long captions 66.1 %, both close to the segmentation result. These results show that semantic content — specifically, which categories of objects appear in each dataset — is a primary source of dataset bias.

A VAE experiment provides further context. Passing images through the bottleneck of a pretrained Stable Diffusion VAE and reconstructing them only reduces accuracy from 82.0 % to 77.4 %, showing that low-level properties such as jpeg compression artefacts or resolution differences contribute relatively little to the overall bias.

The structural transformations produce a surprising result. Canny edge detection achieves 71.0 % accuracy, SAM-based class-agnostic contour extraction reaches 73.2 %, and monocular depth estimation with Depth-Anything-V2 yields 73.1 %. These representations contain no object labels or semantic information, yet they *outperform* most semantic transformations. This shows that the shapes and spatial arrangements of objects differ significantly across datasets and carry as much discriminative information as object category — an important finding for any bias mitigation strategy focused solely on semantic content.

To separate the contribution of local structure from global spatial layout, Zeng et al. run pixel-level and patch-level shuffling experiments. Randomly shuffling individual pixels removes all spatial structure and limits the classifier to colour statistics alone; accuracy falls to 52.2 %, just above the 33.3 % chance level. Shuffling 16×16 patches instead recovers nearly all discriminability, reaching 81.2 %. This shows that local spatial structure within small image regions is sufficient to distinguish datasets, and that global composition matters much less than local texture and shape. A mean-RGB experiment pushes this further by compressing each image to a single average colour per channel. Even this three-number summary achieves 48.5 % accuracy, roughly 15 percentage points above chance. YFCC images are substantially darker on average than CC or DataComp, likely because YFCC contains more outdoor photography captured under varied natural lighting.

Frequency domain analysis shows that bias is present across the full frequency spectrum. Applying high-pass filters (capturing texture and sharp edges) yields 79.2 % accuracy, while low-pass filters (capturing smooth gradients and global structure) yield 70.4 %, both close to the reference. This means that frequency-domain spurious signals exist at multiple scales simultaneously, and that mitigation approaches targeting a single frequency range will not remove the bias entirely.

Finally, combining two transformations together increases accuracy beyond what either achieves alone. For example, combining pixel shuffling with SAM contours reaches 74.2 %, compared to 73.1 % for SAM contours alone. This shows that different types of spurious signals carry partly independent information, which is directly relevant to the multi-transformation approach used in this thesis.

Propagation of Bias into Generative Models

A central question for this thesis is whether biases in training data carry through into the outputs of generative models. Zeng et al. test this by training separate unconditional Diffusion Transformer (DiT) models on each of the three YCD datasets and classifying the resulting synthetic images. These generated images can be classified with 77.6 % accuracy, nearly matching the 82.0 % reference on real images. This demonstrates that generative models reproduce the distributional biases of their training data to a degree that is largely preserved in their outputs. A second experiment using a single pretrained SDXL-Turbo model conditioned on LLaVA-generated captions achieves 58.1 %, confirming that semantic bias is partly transmitted even when the generative model is not dataset-specific. Together, these results show a clear link between training data composition and generative model output, directly supporting the central hypothesis of this thesis.

Semantic Explanation of Dataset Differences

To explain *why* the datasets are distinguishable, Zeng et al. use two complementary approaches. First, Gradient-weighted Class Activation Mapping (Grad-CAM) applied to the reference classifier shows that the model focuses on specific objects – such as elephant herds, tables, and pens – to identify dataset origin. This motivates the use of object-level analysis as an interpretability tool.

A logistic regression model trained on binary object-presence vectors from the ImageNet, LVIS, and ADE20K vocabularies achieves 52.0–52.4 % accuracy depending on the vocabulary, well above the 33.3 % chance level but well below the 82.0 % of the neural classifier. This gap shows that object identity alone does not fully explain the neural classifier’s ability to distinguish datasets, meaning that lower-level and structural signals also play a role. The regression coefficients reveal clear patterns: YFCC is associated with outdoor infrastructure such as traffic lights, poles, and stages, while CC and DataComp are associated with household items, products, and digital graphics. YFCC also contains a notably higher variety of objects per image than the other two datasets, consistent with its origin in less-curated, user-generated Flickr photography.

Open-ended language analysis using LDA topic modelling and GPT-4o summarisation of LLaVA-generated captions characterises YFCC as outdoor and human-centred, DataComp as product-focused with clean backgrounds and few people, and CC as a mix of the two. These findings are validated by prompting LLaVA to answer simple binary questions – such as “Is this image a product showcase?” and “Is this image depicting an outdoor scene?” – across all three datasets, confirming that the identified differences reflect genuine properties of the data rather than quirks of the analysis method.

Gender Artifacts in Visual Datasets

Meister et al. [9] systematically investigate the nature and prevalence of *gender artifacts* in large-scale computer vision datasets. A gender artifact is defined as a visual cue that is both *learnable* — a modern image classifier can use it to distinguish images labelled female from those labelled male — and *interpretable*, meaning it has a recognisable human corollary such as colour, shape, or scene composition. This dual constraint focuses the analysis on the most consequential correlations: those that automated systems can exploit and that carry socially legible meaning.

The work is motivated by a gap in the fairness literature. A widely used family of bias-mitigation techniques, collectively termed *fairness through blindness*, assumes that removing a person’s visible appearance from an image is sufficient to eliminate the gendered signal. Meister et al. challenge this directly by asking a more fundamental question: to what extent can gendered information truly be removed from a visual dataset?

One methodological point is important for interpreting the results. The authors call their model a *gender artifact model* rather than a gender classifier, because it is a diagnostic tool for understanding dataset correlations, not a system intended to predict gender identity. The gender labels used are derived from external annotators’ perceptions rather than self-reported identity, reflecting the limitations of existing dataset annotation practices, and are used solely to reveal distributional differences in the data.

Datasets and Model

The analysis focuses on two widely used object-recognition benchmarks: Microsoft COCO [27] and OpenImages [28]. For COCO, gender labels are inferred from image captions using a keyword-matching heuristic following prior work [26]; for OpenImages, labels come from the More Inclusive Annotations for People (MIAP) subset [29]. Both datasets are skewed male, with 69.2 % and 62.9 % male-labelled images in COCO and OpenImages respectively. The gender artifact model is a ResNet-50 [He_2016_DeepResidualLearning] pretrained on ImageNet, with hyperparameters tuned on a held-out validation set. Performance is reported as the area under the ROC curve (AUC), where 50 % represents chance-level discrimination.

Resolution and Colour Artifacts

The first set of experiments investigates artifacts in the global perceptual properties of images. Images are progressively downsampled from 224×224 to 7×7 pixels, with the model trained and evaluated at each resolution separately. In colour images, the AUC

remains well above chance even at 7×7 (59.2 % on COCO; 63.9 % on OpenImages), a resolution at which almost all semantic content is destroyed. Repeating the experiment in greyscale, the AUC falls to near chance at 28×28 , showing that colour is the primary carrier of gender-correlated signal at low resolutions, and that the model's ability to discriminate cannot be attributed to residual shape or object information.

To isolate this colour artifact in its simplest form, the authors train a logistic regression classifier on just three features: the mean pixel value of the red, green, and blue channels. This achieves AUCs of $58.0\% \pm 0.4$ and $59.1\% \pm 0.4$ on COCO and OpenImages respectively. The likely explanation is a compositional confound: images of male-labelled individuals are disproportionately set in outdoor sports contexts dominated by green tones, while images of female-labelled individuals more often feature close-up portraits with lighter average pixel values.

Person and Background Artifacts

The second set of experiments uses a systematic occlusion protocol to separate gender artifacts arising from the person from those arising from the background. The conditions evaluated are: the full unmodified image (*Full*); the full image with the background masked in black (*Full NoBg*); the image with the person replaced by their segmentation mask (*MaskSegm*); the image with the person replaced by a bounding box (*MaskRect*); and combinations of these with and without the background.

Person appearance. With all person pixels visible, the gender artifact model achieves an AUC of $93.4\% \pm 0.2$ on COCO. Removing the background reduces this only slightly, to $92.7\% \pm 0.2$, confirming that the person's appearance is the dominant signal under normal conditions.

Person shape and pose. When the person's appearance is replaced by a white segmentation mask but the background remains visible (*MaskSegm*), the model still achieves $79.6\% \pm 0.3$ on COCO. The silhouette alone — which still reveals cues such as hair length and clothing outline — carries substantial gendered information even before texture or colour are considered. Removing the background as well (*MaskSegm NoBg*), performance drops only to $74.8\% \pm 0.3$, confirming that silhouette shape and position are meaningful gender artifacts independent of scene context.

Taking this further, the authors train the model on the 17 COCO keypoint locations alone, with no pixel content. This yields an AUC of $64.8\% \pm 0.4$. A normalised pose-only representation — keypoints rescaled to a fixed area and centred to remove size and location information — still achieves $58.0\% \pm 0.0$, establishing body pose as a gender artifact in its own right. Qualitatively, male-associated poses tend to

be smaller and active (e.g. sporting actions), while female-associated poses tend to be larger relative to the frame and static. This is supported quantitatively: the normalised area of female-labelled persons in the COCO training set (0.18 ± 10^{-4}) is significantly larger than for male-labelled persons (0.13 ± 10^{-5}), confirmed at $p < 0.01$ and replicated in OpenImages.

Background as a gender artifact. When the person is occluded but the background is left intact, the gendered signal is not eliminated. Under the MaskRect condition – matching the occlusion strategy used by several prominent mitigation methods [Hendricks_2018_WomenAlsoSnowboard, Wang_2019_BalancedDatasets] – the AUC remains at $70.8\% \pm 0.3$ on COCO and $63.1\% \pm 0.4$ on OpenImages. This shows that the background scene itself encodes gender-correlated information, which directly undermines the core assumption of fairness-through-blindness approaches.

Class Activation Map (CAM) analysis explains why: when the person is occluded, the model shifts its attention onto background objects, associating indoor objects (e.g. ovens and beds) with female labels and outdoor objects (e.g. skateboards and motorcycles) with male labels. Scene type, not just pixel statistics, is the operative artifact in these conditions.

Out-of-distribution evaluations – training on COCO and testing on OpenImages, and vice versa – confirm that these patterns hold across datasets, suggesting they reflect broad regularities in how photographs are composed rather than quirks of any single collection process.

Contextual Object Artifacts

The third set of experiments examines the gender information carried by the identity of objects in a scene. A logistic regression classifier trained on a binary object-presence vector achieves an AUC of 75.4% on COCO and 63.3% on OpenImages. The classifier’s weights reflect familiar stereotypes: objects most predictive of female-labelled images include hair driers, handbags, and teddy bears, while those most predictive of male-labelled images include skateboards, ties, and snowboards.

An iterative removal experiment shows that this gendered information is not concentrated in a few obviously gendered objects. Retaining only the 14 most gendered objects still achieves 72.1% AUC. Random removal tells the same story: 30 randomly selected objects from the 80 annotated COCO categories yield 66.8% , and just 10 randomly chosen objects still achieve 58.0% . Gendered information is distributed broadly across the object vocabulary, and there is no small set of objects whose removal would meaningfully eliminate object-level artifacts.

Analysis of Fairness-Through-Blindness Methods

To evaluate how existing mitigation methods interact with these artifacts in practice, the authors apply the adversarial debiasing approach of Wang et al. [Wang_2019_BalancedDatasets] to COCO images. This method trains a U-Net to identify and remove image pixels that carry salient gender information. When the authors examine which pixels are actually removed from held-out validation images, the results are striking: 76.8 % of removed pixels come from the background, not the person, and $20.4 \% \pm 15.7$ of all background pixels across the dataset are masked out. The objects most frequently affected include giraffes, trains, elephants, buses, and horses — objects commonly found in the active outdoor scenes associated with male-labelled images.

This shows that the method ends up suppressing scene-level information — including objects that may be necessary for downstream tasks such as object recognition — rather than the person-centric gendered information it was designed to target. The broader lesson is that because gendered signals are spread throughout the image, any method trying to remove them will inevitably also remove task-relevant content.

Discussion and Broader Implications

The authors draw three main conclusions. First, gender artifacts are so widely distributed across image structure — from mean colour to silhouette shape to object co-occurrence — that removing them without also removing task-relevant information is not feasible. Second, the experiments reveal an incoherence in what a gender classifier actually predicts: if a model can infer gender labels from the mean pixel colour, a 7×7 blurred image, or the presence of a skateboard, then what it has learned reflects the compositional biases of the dataset rather than gender expression in any meaningful sense. Third, the authors recommend moving away from fairness-through-blindness approaches towards *fairness-aware* methods that explicitly account for distributional differences across demographic groups, and towards disaggregated evaluation practices that make those differences visible.

2.3 Demographic Annotation in Large-Scale Image Datasets

To study demographic bias in large-scale visual datasets and generative models, images must be annotated with sensitive attributes such as gender and skin tone. These annotations are the variable against which spurious feature distributions are measured, so the quality and validity of the annotation process is a central methodological concern. Unlike object category labels, where a clear ground truth can be defined in principle, demographic annotations are contested. They sit at the intersection of biological, social, perceptual, and cultural factors, which makes them inherently harder to assign reliably. This section surveys the annotation practices used in large-scale computer vision research, examines the sources of measurement bias and label uncertainty in both human and automated annotation, reviews the main frameworks used for skin tone annotation, outlines the core limitations of visual demographic inference, and addresses the specific challenges that arise when annotating synthetic images produced by generative models.

2.3.1 Human and Automated Annotation Practices

Demographic attributes in visual datasets have traditionally been labelled by human annotators. Early face recognition datasets such as Labelled Faces in the Wild (LFW) [30] included identity labels and exhibited variation in attributes such as gender and race, but were not designed explicitly for demographic fairness analysis. Large-scale object-centric benchmarks such as MS-COCO [27] include the category “person,” yet do not provide structured demographic annotations. OpenImages [28] extends this hierarchy by including subclasses such as man, woman, boy, and girl, but these annotations were not originally developed under a fairness-driven protocol and are not exhaustive. A more explicit attempt at inclusive demographic annotation is the More Inclusive Annotations for People (MIAP) dataset [29], which revises the OpenImages person hierarchy and introduces non-binary gender presentation labels under a formal multi-annotator framework designed to support fairness analysis. Considered together, these efforts demonstrate that human annotation of demographic attributes is feasible at scale, but methodologically complex. Unlike object category labels, demographic attributes are highly sensitive to annotator subjectivity, cultural background, and perceptual ambiguity, making disagreement and label uncertainty intrinsic rather than incidental to the task.

Inter-annotator agreement for demographic labels is typically lower than for object category labels, reflecting the greater interpretive ambiguity involved in assigning socially constructed attributes. Studies of automated gender classification systems show that performance degrades most sharply for individuals whose gender

presentation does not conform to binary or culturally dominant norms [31]. This pattern suggests that ambiguity and non-conformity pose systematic challenges for both human and machine labeling processes. Crucially, such disagreement cannot be treated as mere annotation noise. Aroyo and Welty [32] argue that annotator disagreement often encodes genuine differences in perception and interpretation, and that collapsing annotations to a single majority label can obscure meaningful variation in how examples are understood.

For skin tone annotation, reliability is further complicated by the fact that visible appearance depends not only on intrinsic pigmentation but also on imaging conditions. Clinical and web-sourced image datasets frequently vary in illumination, camera calibration, and post-processing, making it difficult to disentangle perceived skin tone from lighting artefacts [33]. As a result, both perceptual ambiguity and technical variability contribute to uncertainty in demographic labels.

At the scale of web-scraped datasets such as Re-LAION-5B, manual demographic annotation is not practically feasible. As a result, large-scale studies frequently rely on automated demographic inference, in which pre-trained classifiers are applied to images to estimate attributes such as gender, perceived race, age, or skin tone. Datasets and models such as FairFace [34], which provides race, gender, and age labels across a relatively balanced distribution of seven racial categories, are commonly used for this purpose due to their accessibility and improved demographic coverage relative to earlier benchmarks.

However, automated annotation introduces a structural circularity. Demographic classifiers are themselves trained on human-annotated datasets that encode the same perceptual subjectivities, cultural assumptions, and representational imbalances discussed above. As demonstrated by Buolamwini and Gebru [35], performance disparities in commercial gender classifiers reflect underlying dataset imbalances and phenotypic skew. When such models are subsequently used to annotate new datasets, they propagate the statistical structure and potential biases of their training data into downstream analyses.

Automated demographic inference therefore does not eliminate the epistemic limitations of manual annotation; rather, it relocates them into a learned decision boundary whose internal criteria are opaque. At scale, this can transform annotation bias from an observable human disagreement into a systematic, model-mediated measurement artefact.

2.3.2 Measurement Bias and Label Uncertainty

Measurement bias arises when the process used to assign labels systematically distorts the metric it intends to measure [5]. In the context of demographic annotation, this

distortion can occur at multiple stages: through annotator subjectivity, dataset imbalance, or automated inference error. Unlike object categories, demographic attributes such as gender presentation or skin tone lack stable visual ground truth and are instead inferred from perceptual cues. As a result, measurement error is not purely random but often structured.

Empirical audits of commercial gender classifiers demonstrate that error rates vary substantially across intersectional subgroups. Buolamwini and Gebru [35] show that darker-skinned females experience significantly higher misclassification rates than lighter-skinned males, revealing systematic disparities tied to both observable traits and dataset composition. When automated classifiers trained on such data are used to annotate large-scale web datasets, these subgroup-specific error patterns become embedded in the generated labels.

In large-scale auditing pipelines, confidence thresholds are often used to discard low-probability predictions. While this improves precision, it can also disproportionately exclude images whose appearance deviates from dominant training distributions. Because datasets such as FairFace were explicitly constructed to mitigate racial imbalance [34], their existence highlights the fact that earlier benchmarks were systematically skewed. If annotation models inherit residual imbalances, confidence filtering may amplify rather than eliminate selection effects.

These issues are particularly consequential when demographic labels function not as outcomes but as covariates against which feature distributions are measured. In such settings, structured annotation error can produce apparent correlations between visual artefacts and demographic attributes that reflect classifier error patterns rather than properties of the underlying data. Without ground truth labels at scale, disentangling genuine distributional structure from measurement artefact becomes statistically intractable. Careful selection and documentation of annotation models is therefore essential when demographic attributes are used to audit spurious correlations in large-scale visual datasets.

2.3.3 Skin Tone Annotation Frameworks

The choice of skin tone annotation framework is a methodological decision that directly affects the validity of downstream fairness and correlation analyses. A skin tone scale must satisfy two criteria for use in large-scale visual auditing: (i) construct validity for image-based measurement, and (ii) operational stability under real-world imaging conditions.

The most widely used skin tone framework in computer vision is the Fitzpatrick Skin Type (FST) scale. Originally developed in dermatology to classify sun-reactive skin response to ultraviolet exposure [36], the six-point scale was not designed to measure

visual pigmentation per se, but rather tanning and burning propensity. Its categories are anchored in self-reported photoreactivity rather than directly observable colour properties, creating a construct mismatch when applied to image-based demographic annotation. Subsequent work has documented both the underrepresentation of darker skin types in dermatological datasets and the compression of variation at the darker end of the Fitzpatrick scale [33]. In fairness auditing contexts, such compression reduces sensitivity precisely where subgroup disparities are often most pronounced. The scale's clinical origin and uneven granularity therefore limit its suitability as a measurement instrument for large-scale image datasets.

The Monk Skin Tone (MST) scale was developed to address these limitations by providing a ten-point ordinal framework explicitly designed for technology and machine learning applications [37]. Unlike Fitzpatrick, the MST scale is anchored to visual reference swatches intended for photographic assessment rather than sun-reactive history. Empirical annotation studies show that trained annotators can reliably apply MST categories in alignment with expert judgments, although systematic variation across annotator groups remains observable [37]. This highlights an important methodological point: while MST improves construct alignment for image-based tasks, skin tone annotation remains a subjective perceptual process rather than an objective measurement.

Perceived inclusivity also matters when demographic attributes are used for fairness evaluation. A large-scale survey comparing Fitzpatrick, the Monk scale, and a 40-shade commercial palette found that Fitzpatrick was perceived as significantly less representative, particularly among darker-skinned and historically marginalized participants [38]. Notably, the ten-point MST scale was perceived as comparably inclusive to the much finer-grained commercial palette, suggesting that improved colour distribution and category design can increase representativeness without excessive cognitive burden. These findings support the adoption of MST as a more appropriate perceptual measurement instrument for fairness-oriented annotation.

A separate modelling decision concerns how the MST categories are treated analytically. Although skin tone varies continuously and the MST scale is ordinal, fine-grained ten-class prediction is often unstable in uncontrolled photographic conditions. Empirical work evaluating MST estimation under diverse lighting environments demonstrates substantial degradation in fine-grained classification performance, particularly in intermediate tone ranges [39]. When tones are grouped into broader light, medium, and dark categories, estimation stability improves markedly across lighting conditions. For large-scale web-scraped imagery, where illumination and camera calibration are uncontrolled, a grouped treatment therefore provides greater robustness and reduces boundary-level annotation noise. In the context of this thesis, grouping is not a simplification of the construct but a strategy to reduce measurement

variance that could otherwise obscure or artificially induce spurious correlations.

Another methodological risk arises from shortcut learning during skin tone estimation. When classifiers are trained on full images, models may exploit background, clothing, or contextual artefacts correlated with skin tone labels rather than pigmentation itself. Experimental evidence shows that full-image CNN models achieve higher raw accuracy but rely on contextual features, as revealed by Grad-CAM analyses, whereas models trained on segmented facial skin regions exhibit activations concentrated on skin pixels [40]. This distinction is critical: if annotation models rely on contextual artefacts, any subsequent correlation analysis risks detecting dataset-specific backgrounds rather than genuine pigmentation-related structure. For this reason, skin tone classification in this thesis is applied to detected facial skin regions rather than full images, reducing the likelihood that annotation labels encode spurious environmental signals.

2.3.4 Limitations of Visual Demographic Inference

A fundamental limitation of demographic annotation from visual data is the distinction between perceived attributes and self-identified attributes. Systems that infer gender from images do not measure gender identity; rather, they predict whether a subject's appearance conforms to culturally dominant norms of gendered presentation. Scheuerman et al. demonstrate that commercial gender classification systems operationalise gender as a binary perceptual category and exhibit systematic failures on individuals whose presentation diverges from normative expectations [31]. In most visual datasets, gender labels are derived from annotator perception rather than self-reported identity. Consequently, what is measured is not gender identity but perceived gender expression.

This distinction has methodological implications for fairness evaluation. When generative outputs are compared to demographic statistics—such as labour force participation rates disaggregated by gender—the annotation variable and the statistical benchmark may refer to different constructs. The former reflects appearance-based perception; the latter reflects identity-based self-reporting. Any discrepancy between these constructs introduces ambiguity into downstream bias measurement.

A second limitation concerns the instability of photometric inference under varying acquisition conditions. Skin tone classifiers operate on visible colour values, which are sensitive to illumination colour temperature, exposure, sensor characteristics, and post-processing. In uncontrolled web-scale datasets such as Re-LAION-5B, images are captured under heterogeneous lighting conditions ranging from professional studio setups to low-light mobile photography. Annotation error under such variability is unlikely to be random. Certain lighting styles, aesthetic

conventions, or photographic contexts may be overrepresented in particular communities or content domains, creating a confound between acquisition-level artefacts and demographic inference quality.

This issue is further compounded in synthetic imagery. In diffusion-based models such as Stable Diffusion v1.5, lighting is not an external environmental factor but a learned prior embedded in the generative model. If the model preferentially associates specific lighting distributions with certain prompts or demographic cues, automated skin tone classification applied to generated images may conflate representational bias with generative lighting bias. In this setting, annotation error can become entangled with the model’s learned visual style, making it difficult to separate genuine demographic skew from measurement artefact.

A related conceptual distinction must be drawn between race and skin tone. Race is a socially constructed category shaped by historical, political, and cultural processes; it cannot be reliably inferred from visual appearance alone. Skin tone, by contrast, is a photometric property of visible surface pigmentation. While correlated with racial categorisation in many societies, the two constructs are not equivalent. Fairness audits that operationalise race through perceived visual categorisation introduce additional layers of annotator subjectivity and cultural variability [41]. Skin tone, though still perceptually mediated and affected by imaging conditions, is more directly grounded in observable colour properties and more closely aligned with the low-level image features examined in spurious feature analysis.

Taken together, these limitations highlight that demographic labels derived from visual inference function as measurement instruments rather than ground truth variables. They encode perceptual conventions, acquisition conditions, and annotation design choices. In large-scale visual audits, observed correlations between image features and demographic attributes may therefore reflect properties of the annotation process itself. Recognising this constraint is essential when interpreting spurious feature distributions in both web-scraped datasets and generative model outputs.

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Appendix A Sample A

Appendix B Sample B